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SECTION: **BSSE V-B**

COURSE: **CLOUD COMPUTING**

ASSIGNMENT # 01

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

- Updated Ubuntu packages and installed Docker Engine, curl, and the Docker Compose plugin.
- Started Docker, configured user permissions, and verified Git, Docker, and Docker Compose installations.
- Cloned the provided Gitea setup repository and started Gitea and PostgreSQL using Docker Compose.
- Verified that both containers were running and checked the Gitea HTTP response..

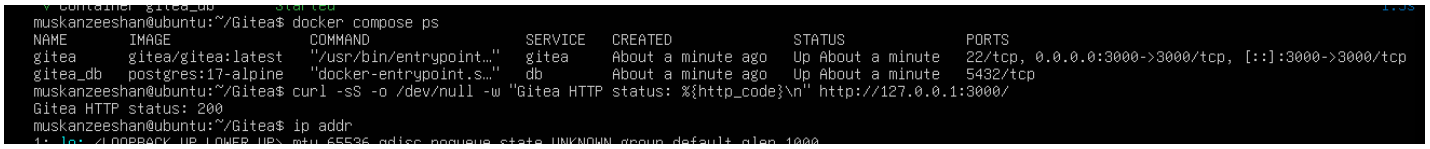


Figure 1: task1_gitea_running.png

Step 1.2: Create and Push the Initial Repository

- Created a public Gitea repository named Assignment01.
- Created a separate local Git repository on the Ubuntu server and switched to the main branch.
- Added my full name and registration number to README.md and committed the file.
- Configured the Gitea remote and successfully pushed the initial commit using my access token.

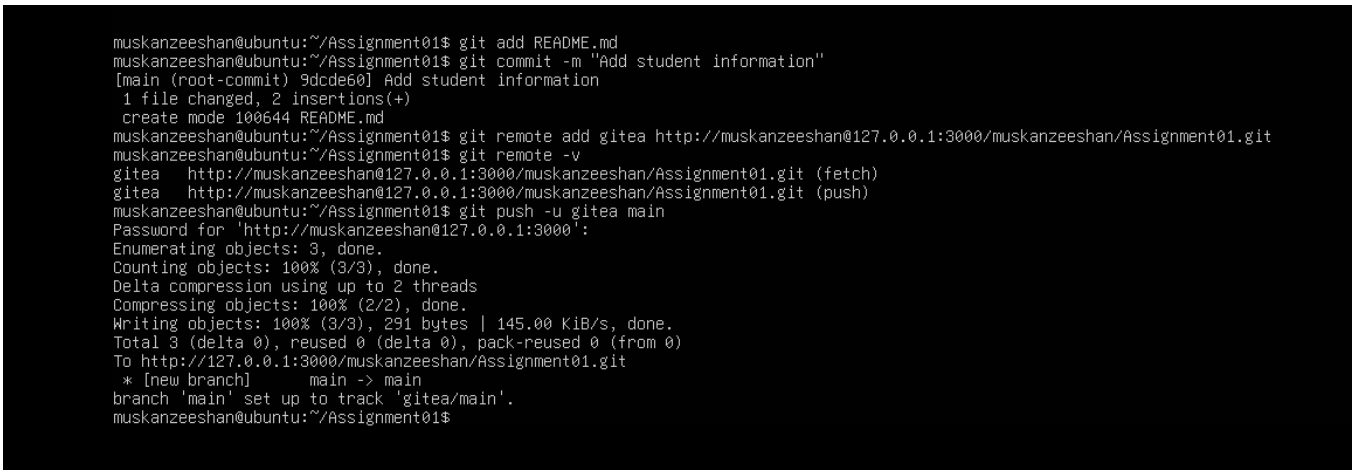


Figure 2: task1_gitea_push.png

Step 1.3: Verify the Gitea Repository

- Opened the Assignment01 repository in Gitea.
- Verified the repository owner and repository name.
- Confirmed that the repository was public.
- Checked that the README displayed my full name and registration number.

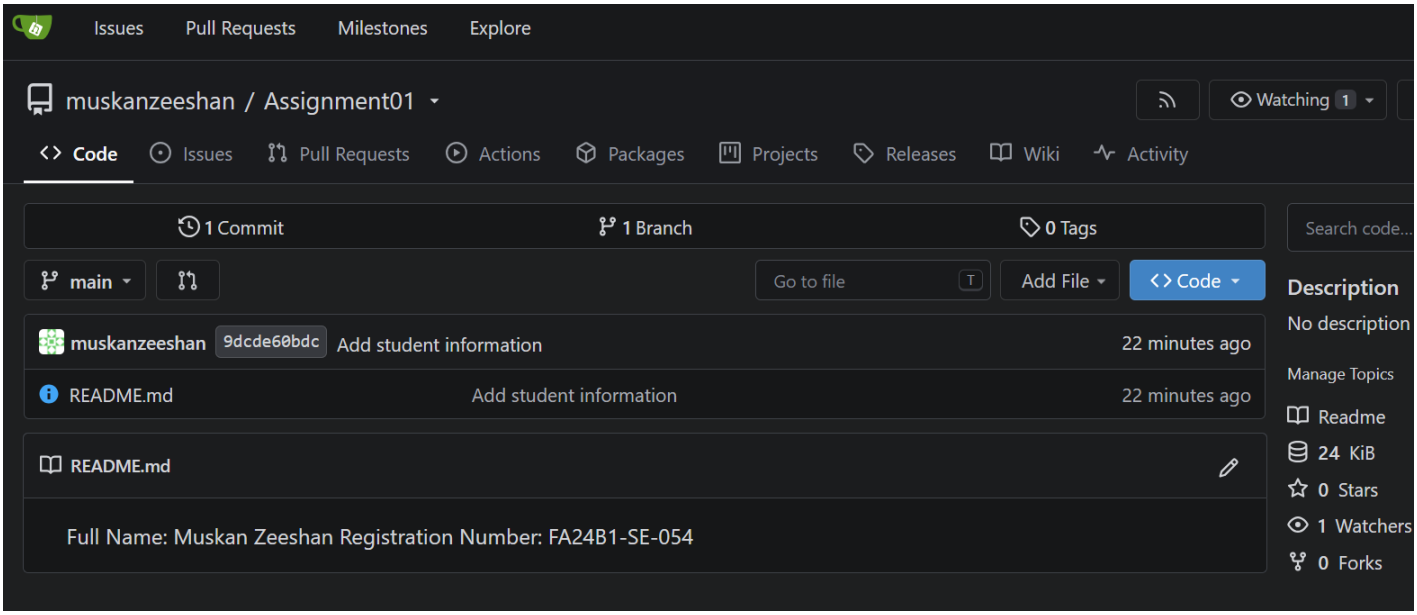


Figure 3: task1_gitea_repository.png

Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

- Created a public GitHub repository named `Assignment01`.
- Connected the existing local repository to GitHub using the `github` remote.
- Verified the `gitea` and `github` remote URLs using `git remote -v`.
- Confirmed that neither remote URL contained a password or access token.

```
muskanzeeshan@ubuntu:~/Assignment01$ cd ~/Assignment01
muskanzeeshan@ubuntu:~/Assignment01$ git remote add github https://github.com/muskanzeeshan/Assignment01.git
muskanzeeshan@ubuntu:~/Assignment01$ git remote -v
gitea   http://muskanzeeshan@127.0.0.1:3000/muskanzeeshan/Assignment01.git (fetch)
gitea   http://muskanzeeshan@127.0.0.1:3000/muskanzeeshan/Assignment01.git (push)
github  https://github.com/muskanzeeshan/Assignment01.git (fetch)
github  https://github.com/muskanzeeshan/Assignment01.git (push)
muskanzeeshan@ubuntu:~/Assignment01$ _
```

Figure 4: task2_remotes.png

Step 2.2: Push to GitHub

- Opened the existing local Assignment01 repository on the Ubuntu server.
- Pushed the main branch to GitHub using git push -u github main.
- Verified that the push completed successfully.
- Confirmed that the local repository was connected to the GitHub repository.

```
muskanzeeshan@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': muskanzeeshan
Password for 'https://muskanzeeshan@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 291 bytes | 72.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/muskanzeeshan/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
muskanzeeshan@ubuntu:~/Assignment01$
```

Figure 5: task2_github_push.png

Step 2.3: Verify the GitHub Repository

- Opened my public GitHub Assignment01 repository in a browser.
- Verified the repository owner and repository name.
- Confirmed that the repository was public.
- Checked that the rendered README displayed my full name and registration number.

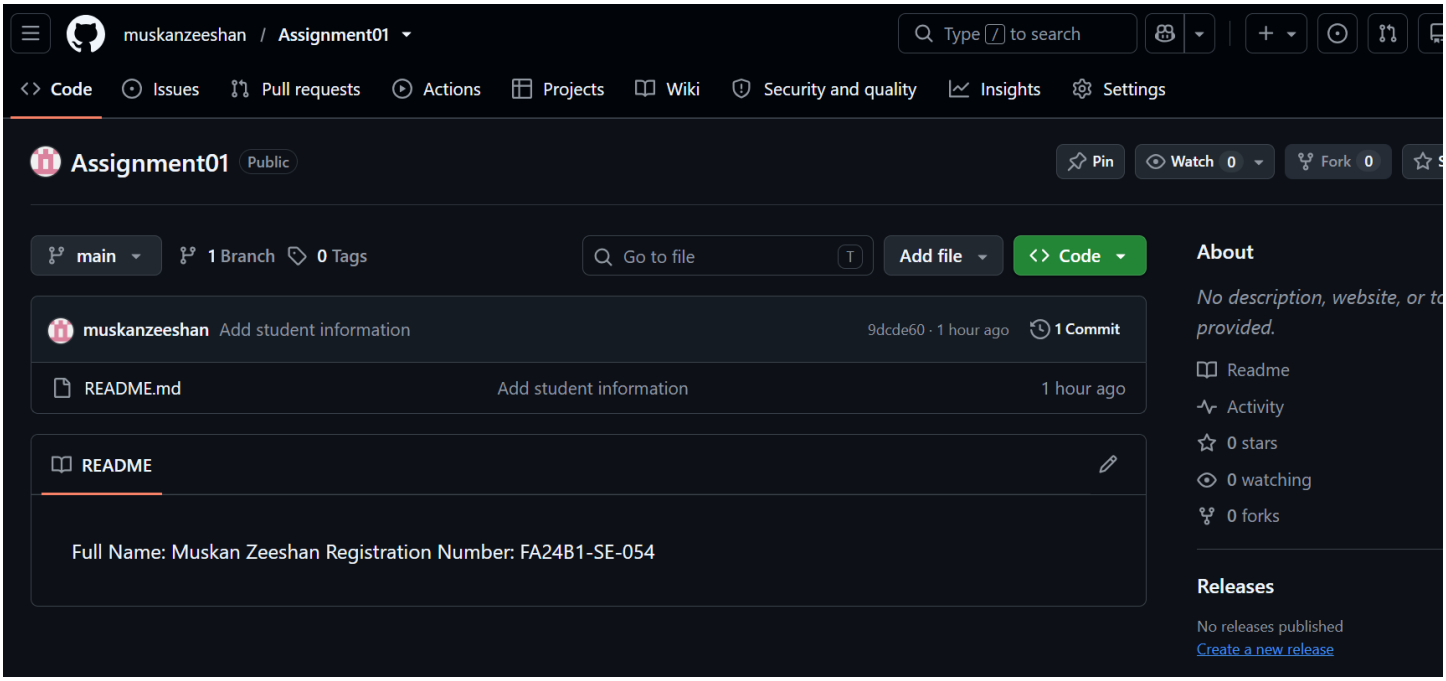


Figure 6: task2_github_repository.png

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

- Opened the local Assignment01 repository on the Ubuntu server.
- Initialized Git LFS using git lfs install.
- Checked the installed Git LFS version.
- Configured Git LFS to track .bin files and verified the rule in .gitattributes.

```
muskanzeeshan@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
muskanzeeshan@ubuntu:~/Assignment01$ cd ~/Assignment01
muskanzeeshan@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
muskanzeeshan@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
muskanzeeshan@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
muskanzeeshan@ubuntu:~/Assignment01$
```

Figure 7: task3_lfs_setup.png

Step 3.2: Create and Verify Three LFS Files

- Created three binary files named large-file-1.bin, large-file-2.bin, and large-file-3.bin.
- Set their sizes to 101 MiB, 102 MiB, and 103 MiB, respectively.
- Added the three files and .gitattributes to Git.
- Used git lfs ls-files --size to verify that all three files were tracked by Git LFS and exceeded 100 MiB.

```
muskanzeeshan@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
muskanzeeshan@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
muskanzeeshan@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
muskanzeeshan@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
muskanzeeshan@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
muskanzeeshan@ubuntu:~/Assignment01$
```

Figure 8: task3_lfs_setup.png

Step 3.3: Commit and Push the LFS Files

- Committed the Git LFS configuration and three large files.
- Pushed the main branch to the GitHub Assignment01 repository.
- Verified that the Git LFS objects uploaded successfully.
- Confirmed that the large files were included in the repository's version-controlled content.

```
muskanzeeshan@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': muskanzeeshan
Password for 'https://muskanzeeshan@github.com':
Username for 'https://github.com': muskanzeeshan
Password for 'https://muskanzeeshan@github.com':
Uploading LFS objects: 100% (3/3), 321 MB | 412 KB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 803 bytes | 803.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/muskanzeeshan/Assignment01.git
   9dcde60..6f019f0  main -> main
muskanzeeshan@ubuntu:~/Assignment01$
```

Figure 9: task3_lfs_files.png

Task 4: Create a Portfolio or CV with GitHub Pages

Step 4.1: Create and Push the Portfolio Repository

- Used my public GitHub portfolio repository named muskanzeeshan.github.io.
- Created or updated the HTML and CSS files for my portfolio/CV.
- Included the required sections: About Me, Education, Skills, and Projects.
- Committed the portfolio source files to the main branch and verified the repository contents.

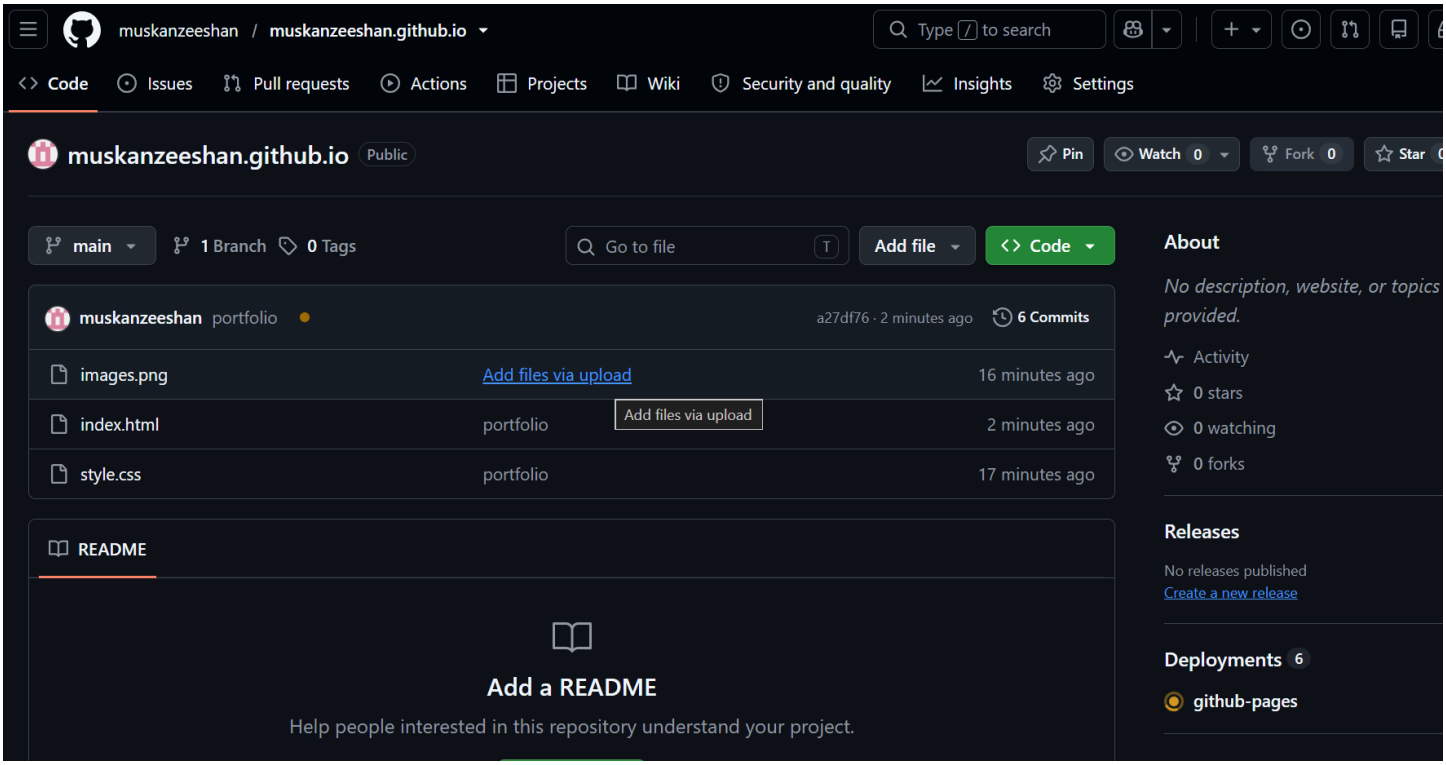


Figure 10: task4_pages_repository.png

Step 4.2: Publish with GitHub Pages

- Opened the portfolio repository settings on GitHub.
- Navigated to Settings → Pages.
- Configured GitHub Pages to deploy from the main branch and the root folder.
- Waited for publication and checked the deployment status and website URL.

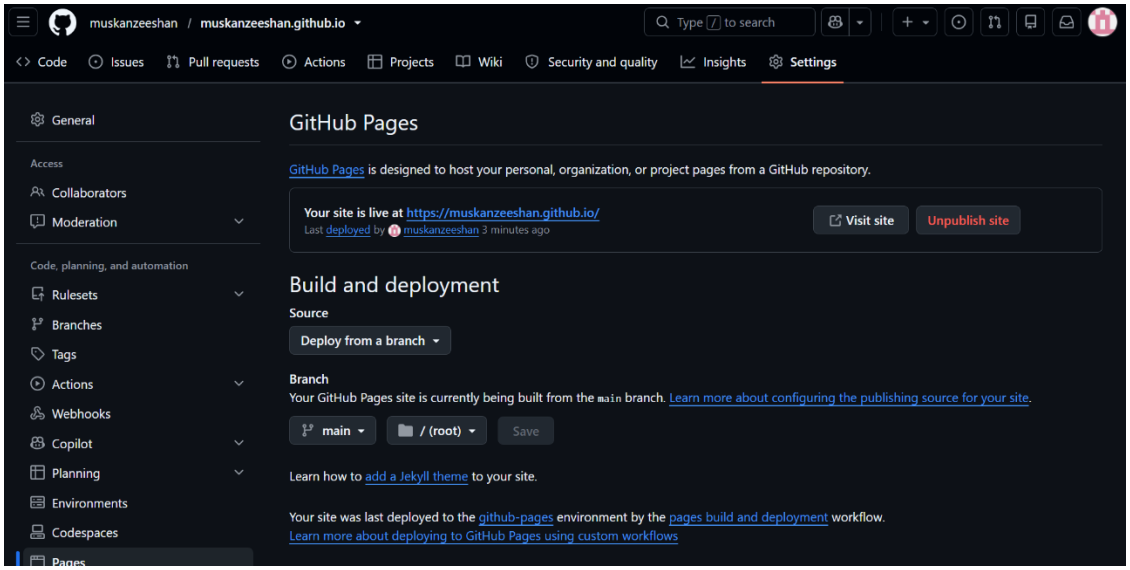


Figure 11: task4_pages_deployment.png

Step 4.3: Verify the Live Portfolio

- Opened https://muskanzeeshan.github.io/ in a web browser.
- Verified that the portfolio website loaded successfully.
- Checked that the portfolio content and required sections were visible.
- Captured a screenshot showing the live website URL and portfolio content.

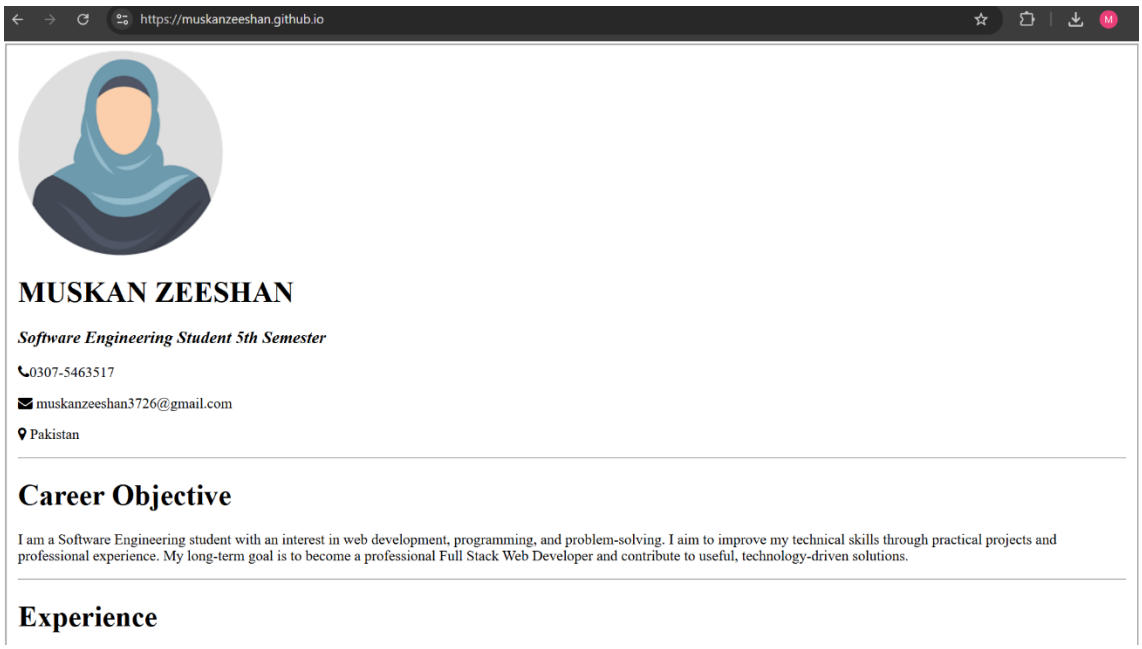


Figure 12: task4_portfolio_live.png